

Processando config 6, seed 3, teste: Rabbit  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Rabbit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

smultin\_MultinomialBitsOver test:

-----

N = 2, n = 8388608, r = 0, s = 32, L = 38,  
Sparse = TRUE

Number of bits = n = 8388608  
Number of cells =  $2^L$  = 274877906944  
Expected number per cell =  $1 / 32768$   
EColl =  $n^2 / (2k)$  = 128  
Hashing = TRUE

Collision test  
CollisionOver: density =  $n / k = 1 / 32768.00$   
Expected number of collisions =  $\mu = 128.00$

-----

Results of CollisionOver test:

POISSON approximation :  
Expected number of collisions =  $N \cdot \mu$  : 256.00  
Observed number of collisions : 268  
p-value of test : 0.23

-----

Total number of cells containing j balls

|       |   |              |
|-------|---|--------------|
| j = 0 | : | 549739036940 |
| j = 1 | : | 16776680     |
| j = 2 | : | 268          |
| j = 3 | : | 0            |
| j = 4 | : | 0            |
| j = 5 | : | 0            |

-----

CPU time used : 00:00:04.47

Generator state:  
16777216 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 262144, r = 0, t = 2

-----  
Max of all bit distances : 17.00  
p-value of test : 0.39

-----  
CPU time used : 00:00:00.05

Generator state:  
33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 131072, r = 0, t = 4

-----  
Max of all bit distances : 8.00  
p-value of test : 0.12

-----  
CPU time used : 00:00:00.08

Generator state:  
50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

svaria\_AppearanceSpacings test:

-----  
N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of  $n = (K + Q)L = 15728640$  bits  
Q = 524288 initialization blocks  
K = 524288 blocks for the test  
the blocks have L = 15 bits

-----  
Normal statistic : 0.60  
p-value of test : 0.27

-----  
CPU time used : 00:00:00.00

Generator state:  
67108864 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

scomp\_LinearComp test:

-----  
N = 1, n = 18176, r = 0, s = 32

-----  
Number of degrees of freedom : 7  
Chi2 statistic for size of jumps : 5.45  
p-value of test : 0.61

-----  
Normal statistic for number of jumps : 0.056  
p-value of test : 0.48

-----  
CPU time used : 00:00:00.18

Generator state:  
67127040 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

scomp\_LempelZiv test:

-----  
N = 1, n = 16777216, r = 0, s = 32, k = 24

-----  
Normal statistic : 0.83  
p-value of test : 0.20

-----  
CPU time used : 00:00:00.40

Generator state:

83904256 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sspectral\_Fourier1 test:

-----

N = 1, n = 1048576, r = 0, s = 32, k = 20

-----

Normal statistic : 0.15  
p-value of test : 0.44

-----

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sspectral\_Fourier3 test:

-----

N = 8065, n = 2048, r = 0, s = 32, k = 11

-----

Kolmogorov-Smirnov+ statistic = D+ : 0.038  
p-value of test : 0.22

Kolmogorov-Smirnov- statistic = D- : 0.022  
p-value of test : 0.60

Anderson-Darling statistic = A2 : 0.63  
p-value of test : 0.62

-----

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sstring\_LongestHeadRun test:

-----  
N = 1, n = 60, r = 0, s = 32, L = 279616

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 2.57  
p-value of test : 0.46

-----  
Global longest run of 1 : 26.00  
p-value of test : 0.12

-----  
CPU time used : 00:00:00.06

Generator state:  
118504992 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sstring\_PeriodsInStrings test:

-----  
N = 1, n = 524288, r = 0, s = 31

-----  
Number of degrees of freedom : 94  
Chi-square statistic : 75.76  
p-value of test : 0.92

-----  
CPU time used : 00:00:00.02

Generator state:  
135282208 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sstring\_HammingWeight test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Number of degrees of freedom : 22  
Chi-square statistic : 19.49  
p-value of test : 0.61  
-----

CPU time used : 00:00:00.00

Generator state:  
152059424 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 0.87  
p-value of test : 0.19  
-----

CPU time used : 00:00:00.00

Generator state:  
168836640 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 64

-----  
Normal statistic : 0.080  
p-value of test : 0.47  
-----

CPU time used : 00:00:00.00

Generator state:  
185613856 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sstring\_HammingCorr test:

-----

N = 1, n = 131072, r = 0, s = 32, L = 128

-----  
Normal statistic : -2.85e-3  
p-value of test : 0.50

-----  
CPU time used : 00:00:00.00

Generator state:  
202391072 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 175.14  
p-value of test : 0.53

-----  
CPU time used : 00:00:00.02

Generator state:  
219168288 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 264.79  
p-value of test : 0.95  
-----

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 489  
Chi-square statistic : 446.25  
p-value of test : 0.92  
-----

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

sstring\_AutoCor test:

-----  
N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :   -1.55
p-value of test          :    0.94
-----
```

```
CPU time used            :   00:00:00.02
```

```
Generator state:
 269499904 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config6_seed3.bin
```

```
sstring_AutoCor test:
```

```
-----
      N =  1,   n = 16777154,   r =  0,   s = 32,   d = 2
-----
```

```
-----
Normal statistic          :    0.11
p-value of test          :    0.46
-----
```

```
CPU time used            :   00:00:00.00
```

```
Generator state:
 286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config6_seed3.bin
```

```
sstring_Run test:
```

```
-----
      N =  1,   n = 3355443,   r =  0,   s =  32
-----
```

```
-----
Total number of 1 runs:  3355444
```

```
Number of degrees of freedom      :   36
Chi2 statistic for number of runs :  38.16
p-value of test                   :    0.37
-----
```

```
-----
Total number of bits:  13425248
```

```
Normal statistic for number of bits :    0.67
p-value of test                   :    0.25
```

-----  
CPU time used : 00:00:00.05

Generator state:  
299702336 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 0.52  
p-value of test : 0.91

-----  
CPU time used : 00:00:00.06

Generator state:  
316479552 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

-----  
Number of degrees of freedom : 2  
Chi-square statistic : 1056.73  
p-value of test : eps \*\*\*\*\*

-----  
CPU time used : 00:00:00.09

Generator state:  
333170752 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 42    |
| ChiSquare statistic          | : | 45.00 |
| p-value of test              | : | 0.35  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 42    |
| ChiSquare statistic          | : | 33.69 |
| p-value of test              | : | 0.82  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 64    |
| ChiSquare statistic          | : | 59.23 |
| p-value of test              | : | 0.65  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 36    |
| ChiSquare statistic          | : | 44.81 |
| p-value of test              | : | 0.15  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 22    |
| ChiSquare statistic          | : | 28.46 |
| p-value of test              | : | 0.16  |

-----  
CPU time used : 00:00:00.06

Generator state:  
349947968 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

-----  
Test on the values of the Statistic H

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 94     |
| ChiSquare statistic          | : | 112.33 |
| p-value of test              | : | 0.10   |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 94    |
| ChiSquare statistic          | : | 84.11 |
| p-value of test              | : | 0.76  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 512    |
| ChiSquare statistic          | : | 523.76 |
| p-value of test              | : | 0.35   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 90    |
| ChiSquare statistic          | : | 96.49 |
| p-value of test              | : | 0.30  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 50    |
| ChiSquare statistic          | : | 46.59 |
| p-value of test              | : | 0.61  |

-----  
CPU time used : 00:00:00.03

Generator state:

366725184 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config6\_seed3.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

-----  
Test on the values of the Statistic H

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 130.70 |
| p-value of test              | : | 0.47   |

-----  
Test on the values of the Statistic M

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 131    |
| ChiSquare statistic          | : | 141.20 |
| p-value of test              | : | 0.26   |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 161    |
| ChiSquare statistic          | : | 169.52 |
| p-value of test              | : | 0.31   |

-----  
Test on the values of the Statistic R

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 124.64 |
| p-value of test              | : | 0.62   |

-----  
Test on the values of the Statistic C

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 89     |
| ChiSquare statistic          | : | 117.61 |
| p-value of test              | : | 0.02   |

-----  
CPU time used : 00:00:00.03

Generator state:

383501984 bits have been read.

===== Summary results of Rabbit =====

|                       |                     |
|-----------------------|---------------------|
| Version:              | TestU01 1.2.3       |
| Generator:            | ufile_CreateReadBin |
| Number of bits:       | 16777216            |
| Number of statistics: | 39                  |

Total CPU time: 00:00:06.13  
The following tests gave p-values outside [0.001, 0.9990]:  
(eps means a value < 1.0e-300):  
(eps1 means a value < 1.0e-15):

| Test                     | p-value |
|--------------------------|---------|
| -----                    | -----   |
| 22 MatrixRank, 320 x 320 | eps     |
| -----                    | -----   |

All other tests were passed